

PYTHON PYQ (PREVIOUS YEAR QUESTIONS) – UNIT WISE

UNIT 1		
S. No.	SECTION A	Year
1.	What type of Language is Python?	2019-20
2.	What are local variables and global variables in Python?	2019-20
3.	Which of the following statements produces an error in Python? x, y, z = 1,2,3 #s1 a, b = 4,5,6 #s2 u = 7,8,9, #s3 (List all the statements that have error.)	2020-21
4.	Explain the role of precedence with an example.	2020-21
5.	Consider the program: x = ['12', 'hello', 456] x[0] *= 3 x[1][1] = 'bye' Explain why this program generates an error.	2020-21
6.	Discuss why Python is called as dynamic and strongly typed language?	2021-22
7.	Define the programming cycle for Python.	2022-23
8.	Explain mutable sequences in Python.	2022-23
9.	Discuss the special methods <code>__le__</code> and <code>__ne__</code> .	2022-23
10.	Explain the output of following function. def printalpha (abc_list, num_list): for char in abc_list: for num in num_list: print(char, num) return printalpha (['a', 'b', 'c'], [1, 2, 3])	2022-23
11.	Write python function to perform linear search.	2022-23
12.	Give difference between <code>==</code> and <code>is</code> operator.	2023-24
13.	Demonstrate how to assign a single value to a tuple.	2024-25

UNIT 1		
S. No.	SECTION B	Year
1.	What is Python? How Python is interpreted? What are the tools that help to find bugs or perform static analysis? What are Python decorators?	2019-20
2.	Compare list and tuple data structure with suitable examples. Explain the concept of list comprehension.	2022-23
3.	Explain the python implementation of stack ADT.	2022-23
4.	Design a basic calculator in Python that supports addition, subtraction, multiplication, division.	2024-25

UNIT 1		
S. No.	SECTION C	Year
1.	Write short notes with example: The Programming Cycle for python. Elements Python, Type Conversion in Python, Operator Precedence, and Boolean Expression.	2019-20
2.	How memory is managed in Python? Explain PEP 8. Write a Python program to print even length words in a string.	2019-20
3.	Explain why Python is considered an interpreted language.	2020-21
4.	What is short circuit evaluations? What Is printed by the following Python Program? a = 0 b = 2 c = 3 x = c or a print(x)	2020-21
5.	What do you mean by operator precedence and associativity. Explain.	2021-22
6.	Write short notes on: i. The programming cycle for python ii. Type conversion in Python	2021-22
7.	Write Python program to swap two numbers without using Intermediate/ Temporary variables. Prompt the user for input.	2021-22
8.	Discuss various categories of operators in python. Find and explain stepwise solution of following expressions if a=3, b=5, c=10. i) $a \& b \ll 2 // 5 ** 2 + c ^ b$ ii) $b >> a ** 2 \ll 2 >> b ** 2 ^ c ** 3$	2022-23
9.	Discuss why python is interpreted language. Explain history and features of python while comparing python version 2 and 3.	2022-23
10.	Write short note on the following with examples: a) Operator precedence b) Python Indentation c) Type Conversion	2024-25
11.	Write a program to validate email addresses using regular expressions. Criteria: 1. Must contain @ symbol 2. Must contain domain name 3. Should not have spaces	2024-25

UNIT 2		
S. No.	SECTION A	Year
1.	In some languages, every statement ends with a semi-colon (;). What happens if you put a semi-colon at the end of a Python statement?	2019-20
2.	How is Python an interpreted language?	2019-20
3.	How pass statement is different from a comment?	2021-22
4.	Write a for loop that prints numbers from 0 to 57, using range function.	2021-22
5.	Discuss the purpose of the break and continue statement in loops.	2022-23
6.	Explain the output of following function. <pre>def printalpha (abc_list, num_list): for char in abc_list: for num in num_list: print(char, num) return printalpha (['a', 'b', 'c'], [1, 2, 3])</pre>	2022-23
7.	Write python function to perform linear search.	2022-23
8.	Can you use else with a for loop? When is it executed?	2023-24
9.	What will be the output of the following Python code? <pre>l=[1,0,0,2,'hi','',[[]]] print(list(filter(bool,l))</pre>	2023-24
10	What will be the output of the following Python code? <pre>def compute(x): return [i**2 for I in x if i%2==0] print(compute([1,2,3,4,5]))</pre>	2024-25
11.	Explain floor division with an example.	2024-25

UNIT 2		
S. No.	SECTION B	Year
1.	Explain all the Conditional statement in Python using small code example.	2019-20
2.	Write a program factors(N) that returns a list of all positive divisions of N (N>=1). For example: factors (6) returns [1,2,3,6] factors (1) returns [1] factors (13) returns [1,13]	2020-21
3.	Explain the use of break and continue with a suitable example	2021-22
4.	Write Python code snippet to display n terms of Fibonacci series using recursion.	2022-23
5.	Write and explain an algorithm through python code to generate prime numbers.	2022-23
6.	Define Membership and Identity Operators. Given: a=3 b=3 Distinguish between: (a is b) and (a==b)?	2024-25

UNIT 2		
S. No.	SECTION C	Year
1.	Explain Expression Evaluation & Float Representation with example. Write a Python Program for How to check if a given number is Fibonacci number.	2019-20
2.	Explain the purpose and working of loops. Discuss Break and continue with example. Write a Python program to convert time from 12 hour to 24 hour format.	2019-20
3.	Write a function makePairs that takes as input two lists of equal length and returns a single list of same length where k-th element is the pair of k-th elements from the input lists. For example. makePairs([1,3,5,7],[2,4,6,8]) returns [(1,2),(3,4),(5,6),(7,8)] makePairs([],[]) retruns []	2020-21
4.	Write a program to check an input number is prime or not.	2021-22
5.	Write a program that accepts a sentence and calculate the number of digits, uppercase and lowercase letters.	2021-22
6.	Write a program to check an input year is leap year or not.	2021-22
7.	Write a python program to count the vowels present in given input string. Explain the output of program through example.	2022-23
8.	Write a program that accepts sequence of lines as input and prints the lines after making all characters in the sentence capitalized. e.g. If Input: Hello world Practice makes perfect Then, Output: HELLO WORLD PRACTICE MAKES PERFECT	2022-23
9.	Write a program to create a hollow pyramid pattern given below: * * * * * * * *****	2024-25
10.	Explain the why loops are needed and the types of loops in python. Discuss break and continue with example.	2024-25

UNIT 3		
S. No.	SECTION A	Year
1.	What is the difference between list and tuples in Python?	2019-20
2.	What is the difference between Python Arrays and lists?	2019-20
3.	Write a recursive Python function “rprint” to print all elements in a list in reverse. rprint([]) prints nothing rprint([1,2,3]) prints 3 2 1	2020-21
4.	Describe the behavior of “range (s,e)” in Python.	2020-21
5.	What is the difference between Python Arrays and Lists?	2021-22
6.	Describe the use of class ` init ()` method.	2022-23
7.	Explain mutable sequences in Python.	2022-23
8.	Describe the purpose of the split() method in string manipulation.	2023-24
9.	Briefly describe the use of lambda functions in Python.	2024-25

UNIT 3		
S. No.	SECTION B	Year
1.	Discuss function in python with its parts and scope. Explain with example. (Take Simple calculator with add, subtract, division and multiplication).	2019-20
2.	Write a Python function average to compute the average of a list of numbers. The function must use try-except to handle the case where the input list is empty. Further, in that case the average for the empty list should be set to 0.0 using the except block.	2020-21
3.	Write a python function named ComputeAverage to find average of a list of numbers. It should handle the exception if the list is empty and return 0 in that case.	2021-22
4.	Compare list and tuple data structure with suitable examples. Explain the concept of list comprehension.	2022-23
5.	Explain the python implementation of stack ADT.	2022-23
6.	Describe python program to implement selection sort. Simulate your code on the following data 2,52,1,5,2,54,62,64.	2022-23
7.	Explain how to define a list in Python. Write a Python program to remove duplicates from a list and print the resulting list.	2023-24
8.	Explain the concept of functions in Python. Write a function that takes a list of numbers and returns the sum of all the numbers in the list.	2023-24
9.	Write Python function to count the frequency of each character in a given string and return the output in a dictionary. Example: char frequency('HELLO') returns {"H":1,"E":1,"L":2,"O":1}	2024-25

UNIT 3		
S. No.	SECTION C	Year
1.	Explain high order function with respect to lambda expression. Write a Python code to Count occurrences of an element in a list.	2019-20
2.	Explain Unpacking Sequences, Mutable, Sequences, and List Comprehension with example. Write a program to sort list of dictionaries by values in Python – Using lambda function.	2019-20
3.	Write a Python program, triangle(N), that prints a right triangle having base and height consisting of N * symbols as shown in these examples: triangle(3) prints: <pre>* ** ***</pre> triangle(5) prints: <pre>* ** *** **** *****</pre>	2020-21
4.	Write a Python function searchMany(s,x,k), that takes as argument a sequence s and integers x,k (k>0). The function returns True if there are at most k occurrences of x in s. Otherwise it returns False. For example: searchMany([10, 17, 15, 12], 15, 1) returns True searchMany([10, 12, 12, 12], 12, 2) returns False searchMany([10, 12, 15, 11], 17, 18) returns True	2020-21
5.	Write a python program to count the vowels present in given input string. Explain the output of program through example.	2022-23
6.	Write a program that accepts sequence of lines as input and prints the lines after making all characters in the sentence capitalized. e.g. If Input: Hello world Practice makes perfect Then, Output: HELLO WORLD PRACTICE MAKES PERFECT	2022-23
7.	Discuss list data structure of python. Explain various inbuilt methods of list with suitable example of each.	2022-23
8.	Compare list and dictionary data structure. Explain various dictionary methods with suitable examples of each.	2022-23
9.	Explain the Tower of Hanoi problem and its recursive solution in Python.	2022-23
10.	Write a function to find the longest word in a given list of words. Example: longest_word(['apple','banana','cherry']) returns 'banana'.	2024-25

11.	Distinguish between a Tuple and List with examples. Explain with examples at least 4 built-in methods of Dictionary.	2024-25
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UNIT 4		
S. No.	SECTION A	Year
1.	Mention five benefits of using Python.	2019-20
2.	Define ADT interface.	2019-20
3.	Discuss various file opening modes of python.	2022-23
4.	Compare assert, try-except and raise statements.	2022-23
5.	What does the readline() function return when it reaches the end of a file?	2023-24
6.	State how to handle exceptions In Python? Provide a simple example.	2024-25
7.	Describe the purpose of the 'with' statement in file handling?	2024-25

UNIT 4		
S. No.	SECTION B	Year
1.	Discuss ADT in python. How to define ADT? Write code for a student information.	2019-20
2.	Write a function lessthan(lst,k) to return list of numbers less than k from a list lst. The function must use list comprehension. Example: lessthan([1,-2,0,5,-3],0) returns [-2,-3]	2020-21
3.	Write a Python program to read a file named "ABC.txt" and count the number of lines, words, and characters in the file.	2023-24
4.	Write a program to reverse the contents of a file character by character, separating each character with a comma.	2024-25

UNIT 4		
S. No.	SECTION C	Year
1.	Discuss File I/O in python. How to perform open, read, write and close into a file? Write a Python to read line-by-line store it into a variable.	2019-20
2.	Discuss Exceptions and Assertions in python. How to handle Exceptions with Try-Finally? Explain 5 built-in exceptions with example.	2019-20
3.	Show an example where both keyword arguments and default arguments are used for the same function in a cell. Show both the definition of the function and its call.	2020-21
4.	Write a python program, countSquares(N), that returns the count of perfect squares less than or equal to N (N>1). For example: CountSquares(1) returns 1 #Only 1 is a perfect square <=1 CountSquares(5) returns 2 #1, 4 <=5 CountSquares(55) returns 7 #1, 4, 9, 16, 25, 36, 49 <=55	2020-21

5.	Write a Python function, <code>alternating(lst)</code> , that takes as argument a sequence <code>lst</code> . The function returns <code>True</code> if the elements in <code>lst</code> are alternately odd and even, starting with an even number. Otherwise it returns <code>False</code> . For example: <code>alternating([10, 9, 9, 6])</code> returns <code>False</code> <code>alternating([10, 15, 8])</code> returns <code>True</code> <code>alternating([10])</code> returns <code>True</code> <code>alternating([])</code> returns <code>True</code> <code>alternating([15, 10, 9])</code> returns <code>False</code>	2020-21
6.	There is a file named <code>Input.Txt</code> . Enter some positive numbers into the file named <code>Input.Txt</code> . Read the contents of the file and if it is an odd number write it to <code>ODD.TXT</code> and if the number is even, write it to <code>EVEN.TXT</code>	2021-22
7.	Discuss Exceptions and Assertions in Python. Explain with a suitable example. Explain any two built-in exceptions.	2021-22
8.	What is the significance of module in Python? What are the methods of importing a module? Explain with suitable example	2021-22
9.	What are different types of inheritance supported by Python? Explain.	2021-22
10.	Discuss the concept of Iterators in python. Explain, how we can create text file in python? Describe python program to write the number of letters and digits in given input string into a File object.	2022-23
11.	Explain different file opening modes also write a Python Program to read a file and capitalize the first letter of every word in the file.	2023-24
12.	Discuss different types of file modes in Python? Explain with examples.	2024-25
13.	Write a program to read a CSV file and display the rows where a specific column value exceeds a given threshold.	2024-25

UNIT 5		
S. No.	SECTION A	Year
1.	Define floor division with example.	2019-20
2.	What is the use of “raise” statement? Describe with an example.	2020-21
3.	Explain the use of “with” construct in Python with an example program.	2020-21
4.	How do you read an input from a user in Python to be used as an integer in the rest of the program? Explain with an example.	2020-21
5.	What is the output of the following program? (lambda x,y : y - 2*x) (1, 11)	2020-21
6.	Explain the use of <code>__init__</code> function in a class Python?	2020-21
7.	Explain features of any two Python IDEs.	2022-23
8.	Which function is used to create identity matrix in NumPy?	2023-24
9.	Explain why numpy is used instead of python arrays for mathematical calculations.	2024-25

UNIT 5														
S. No.	SECTION B	Year												
1.	Explain iterator. Write a program to demonstrate the tower of Hanoi using function.	2019-20												
2.	Write a Python function removekth(s, k) that takes as input a string s and an integer k>=0 and removes the character at index k. If k is beyond the length of s, the whole of s is returned. For example, removekth("PYTHON", 1) returns "PTHON" removekth("PYTHON", 3) returns "PYTON" removekth("PYTHON", 0) returns "YTHON" removekth("PYTHON", 20) returns "PYTHON"	2020-21												
3.	Describe the differences between a linear search and a binary search?	2020-21												
4.	Implement the binary search technique.	2021-22												
5.	Explain the basic usage of matplotlib for plotting graphs. Write a Python program to plot a simple line graph showing the relationship between x = [1,2,3,4,5] and y = [1,4,9,16,25].	2023-24												
6.	Create a pie chart using matplotlib to represent the following data: <table><tr><th>Languages</th><th>Popularity</th></tr><tr><td>Python</td><td>30</td></tr><tr><td>Java</td><td>25</td></tr><tr><td>C++</td><td>20</td></tr><tr><td>JavaScript</td><td>15</td></tr><tr><td>Ruby</td><td>10</td></tr></table>	Languages	Popularity	Python	30	Java	25	C++	20	JavaScript	15	Ruby	10	2024-25
Languages	Popularity													
Python	30													
Java	25													
C++	20													
JavaScript	15													
Ruby	10													

UNIT 5		
S. No.	SECTION C	Year
1.	Discuss and differentiate Iterators & Recursion. Write a program for Recursive Fibonacci series.	2019-20
2.	Discuss Sorting & Merging. Explain different types of sorting with example. Write a Python Program for Sieve of Eratosthenes.	2019-20
3.	How can you create Python file that can be imported as a library as well as run as a standalone script?	2020-21
4.	Describe the difference between import library and from library import * when used in a python program. Here library is some python library.	2020-21
5.	What do you mean by recursion? Write a recursive function to compute the factorial of an input number N.	2021-22
6.	Discuss Exceptions and Assertions in Python. Explain with a suitable example. Explain any two built-in exceptions	2021-22
7.	Illustrate through examples, how we can implement different type of inheritance in python? Explain the static, instance and class method through suitable example.	2022-23
8.	Write python program to implement merge sort algorithm. Discuss, how we can compare the time complexity of two algorithms?	2022-23
9.	Describe how to generate random numbers using NumPy. Write a Python program to create an array of 5 random integers between 10 and 50	2023-24
10.	Explain the concept of DataFrame in pandas. Write a python program to create a DataFrame from a dictionary and print it.	2023-24
11.	Discuss the role of event handling in Tkinter. How can you bind events to widgets? Provide examples.	2024-25
12.	Write a program to read data from a CSV file 'students.csv', calculate the average marks for each student, and display the results.	2024-25